

Table 2: CGCNN Evaluation Metrics Across Train, Validation, and Test Partitions

Property	Unit	Train MAE	Train R2	Val MAE	Val R2	Test MAE	Test RMSE	Test R2
band_gap	eV	0.168	0.973	0.188	0.943	0.211	0.354	0.943
formation_energy	eV/atom	0.139	0.968	0.098	0.945	0.213	0.262	0.938
energy_hull	eV/atom	0.129	0.971	0.157	0.954	0.252	0.299	0.937
bulk_modulus	GPa	9.640	0.966	11.846	0.937	4.967	6.147	0.922
shear_modulus	GPa	7.816	0.969	5.206	0.942	1.924	2.495	0.925
young_modulus	GPa	16.892	0.971	14.197	0.949	5.316	5.924	0.926
E_ads_DFT_eV	eV	0.146	0.966	0.237	0.940	0.189	0.236	0.932